

Soul Stone

Technical Sheet — Participant Guide

Data Space Club — ENSSEA

Coursera Course Reviews Dataset

Quick Summary

Macro F1-Score: 70% | **Code Quality:** 15% | **Report:** 15%

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1 The Problem

You are given a dataset of course reviews written by students. Your task is to **build a model that predicts the rating (1 to 5) based only on the review text.**

This is a **multi-class text classification problem**:

- **Input:** A text review (in English).
- **Output:** A rating from 1 to 5.

2 Files Provided

File	Description
`soul_train.csv`	Training data — contains `Id`, `Review`, `Rate`
`soul_test.csv`	Test data — contains `Id`, `Review` (**`Rate` is hidden**)
`sample_submission.csv`	Submission format template
`technical_sheet.md`	This guide

3 The Dataset

Info

Source: Real Coursera course reviews from students around the world.
Target: `Rate` (1 to 5). The dataset is **imbalanced**.

Column	Type	Description
`Id`	int	Unique identifier for each review
`Review`	str	The full text of the student's review
`Rate`	int	The rating given by the student (**1–5**). **This is your target.**

Important Note

`Rate` is **removed from `soul_test.csv`**. You must predict it.

4 Evaluation Metric

Your submission will be scored using the **Macro F1-Score**:

$$F1_{\text{macro}} = \frac{1}{5}\sum_{i=1}^{5} F1_i$$

Info

Why Macro F1? It treats all classes equally, making it robust to class imbalance. A model that only predicts "5" will get a very low score. **Maximum score = 1.0**

5 Strict Rules — Read Carefully

Important Note

Using Deep Learning = CHEATING = DISQUALIFICATION. This challenge tests your ability to work with **classical NLP and classical Machine Learning**.

Forbidden Techniques

#	Forbidden Technique
1	Deep Learning models (Neural Networks, MLP, DNN)
2	Recurrent Neural Networks (RNN, LSTM, GRU)
3	Convolutional Neural Networks (1D CNN for text)
4	Transformers (BERT, GPT, RoBERTa, DistilBERT, T5, etc.)
5	Pre-trained language models of any kind
6	Word embeddings from neural models (Word2Vec, GloVe, FastText)
7	Any deep learning framework (TensorFlow, PyTorch, Keras)
8	AI-as-a-service APIs (OpenAI, Claude, Gemini, etc.)

6 Allowed Techniques

Allowed

You must use classical approaches. The following are allowed and encouraged.

Text Preprocessing

- Tokenization
- Lowercasing
- Removing stopwords & punctuation
- Stemming / Lemmatization

Text Representation

- Bag of Words
- TF-IDF
- N-grams
- Lexicon features (VADER)

Classical ML Models

- Naive Bayes
- Logistic Regression
- SVM (LinearSVC)
- Random Forest / XGBoost
- K-Nearest Neighbors

Other Practices

- Cross-validation
- Hyperparameter tuning
- Ensemble methods
- Handling imbalance (SMOTE)

7 Submission Format

Submit a CSV file named `submission.csv`.

Column	Type	Description
`Id`	int	Must match the test set exactly.
`Rate`	int	Integer between 1 and 5.

```
Id,Rate 1,5 2,3 3,1 ...
```

Important Note

Do **not** submit probabilities. Submit the final integer class.

8 Suggested Workflow

Phase	Task	Time
1	Load data, explore distribution	30 min
2	Text preprocessing	1 hour
3	Text representation (TF-IDF)	30 min
4	Train baseline model	30 min
5	Try multiple models	1 hour
6	Hyperparameter tuning	1 hour
7	Handle class imbalance	30 min
8	Final predictions + submission	30 min
Total		**5.5h**

9 Submission Checklist

- ☑ No Deep Learning model used.
- ☑ No pre-trained language models (BERT, GPT).
- ☑ No pre-trained word embeddings.
- ☑ `submission.csv` has the exact format.
- ☑ `Id` values match the test set.
- ☑ `Rate` values are integers between 1 and 5.
- ☑ 3 files submitted:
  - `submission.csv`
  - `notebook.ipynb` or `script.py` (Judges check for DL)
  - `report.pdf` (1–2 pages)

⚠ Important Note

If your code is not submitted, your team will be **disqualified**. Judges must verify that no Deep Learning was used.

10 Evaluation Summary

Component	Weight
Macro F1-Score on test set	70%
Code quality + clarity	15%
Report quality	15%